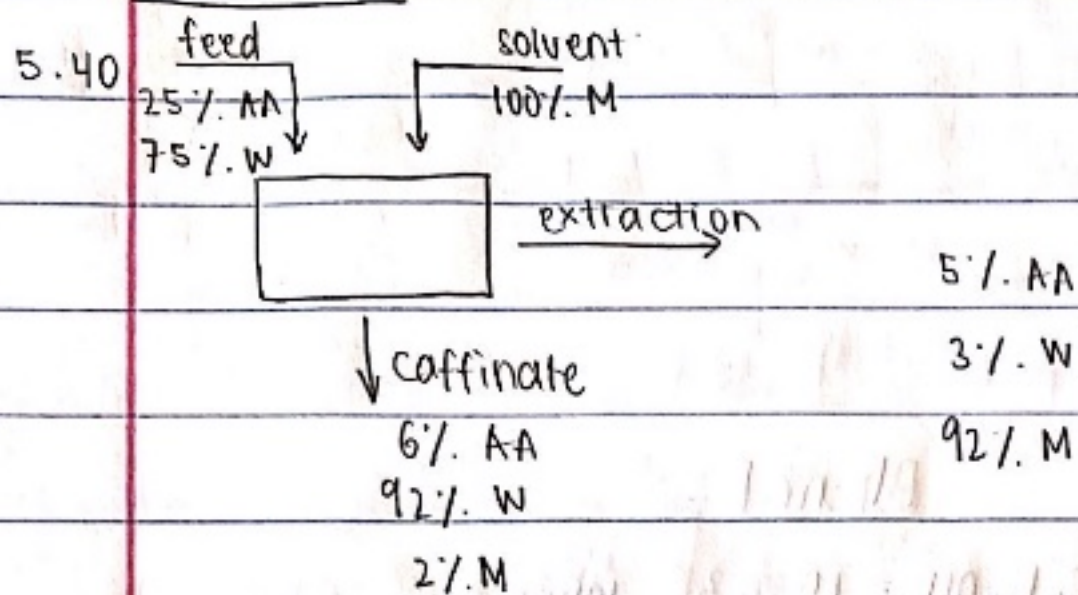


49
53

HW #9



$$\text{Material Balance} \quad 0.05 M_E = \frac{25 - 0.06 M_R}{0.05}$$

$$25 = 0.06 M_R + 0.05 M_E \quad 500 = 1.2 M_R$$

$$75 = 0.92 M_R + 0.03 M_E \quad 75 = 0.92 M_R + 0.05(300 - 12 M_R)$$

$$\dot{m}_S = 0.02 M_R + 0.92 M_E = 0.92 M_R + 15 - 0.036 M_R$$

$$= 0.02(67.87) + 0.92(418.556) = 0.42 M_R$$

$$= 3864.2 \text{ kg/hr}$$

$$\dot{m}_S + \dot{m}_F = \dot{m}_F + \dot{m}_R$$

$$386.42 + 100 = 418.556 + 67.87$$

$$486.42 = 486.426$$

$$75 = 0.884 M_R + 15$$

$$60 = 0.884 M_R$$

$$M_R = 67.87 \text{ kg/hr}$$

$$25 = 0.06(67.87) + 0.05 M_E$$

$$\frac{20.4278}{0.05} = -4.0722 \quad \frac{0.05 M_E}{0.05}$$

$$M_E = 418.556 \text{ kg/hr}$$

We will need 3.8642 kg per kg of feed

$$3. a). n = \frac{P V_{out}}{R T_{out}} = \frac{122 \times 10^3 \text{ Pa} \cdot 1.25 \frac{\text{m}^3}{\text{min}}}{8.314 \frac{\text{m}^3 \text{Pa}}{\text{kmol}} \times (150 + 273.15) \text{ K}} \cdot \frac{60 \frac{\text{sec}}{\text{min}}}{3541.6667 \frac{\text{mol}}{\text{sec}}}$$

$$n = 0.7224 \frac{\text{mol}}{\text{sec}}$$

Energy Balance Eq.

$$\Delta H + \Delta PE + \Delta KE = Q - W$$

$$\Delta H = Q$$

ΔPE & ΔKE = negative

$$\Rightarrow \Delta H = 0.722 \frac{\text{mol}}{\text{sec}} \times 0.264 \frac{\text{J}}{\text{mol}}$$

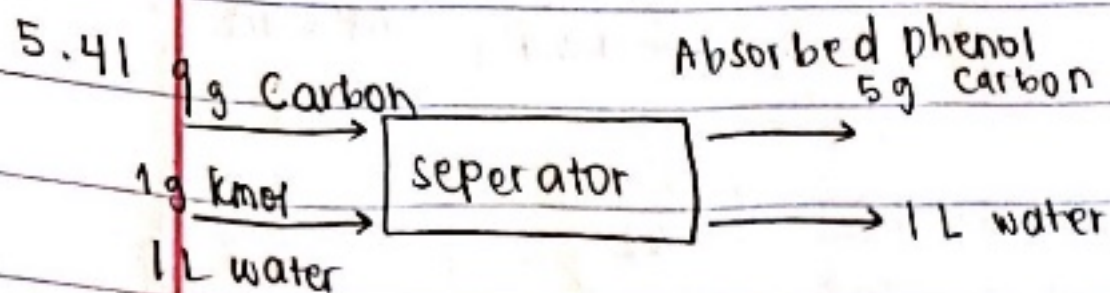
no shaft work ($W=0$)

$$Q = \Delta H = 2628.08$$

$$= 2628.08 \text{ W} \left(\frac{1}{1000} = \frac{\text{kW}}{\text{W}} \right)$$

$$Q = 2.628 \text{ kW}$$

b) To calculate ΔKE , we would use $\frac{3}{2} k_B \Delta T$ (some would need Boltzmann constant). Since temperature would be rising, the value of ΔT will be positive. This will make our kinetic energy positive.



Test values was curve

Phenol

Ca	qa	Absorbed %
0	0	0.00
0.1	0.8	38.54
0.2	1.25	60.63
0.5	1.95	96.35
0.6	2.0	99.68
0.7	2.175	108.8

$$C_6H_5OH = 10.638 \text{ mmol}$$

$$C_p = \text{phenol concentration (mmol/L)}$$

$$q_e = \text{phenol absorber (mmol/g)}$$

$$10.638 = C_p(1L) + q_p(5g)$$

$$\% \text{ absorbed} = \frac{C_p(1L) + q_p(5g)}{10.638 \text{ mmol}}$$

$$10.638 \text{ mmol}$$

$$\text{phenol in water} = 0.6 \text{ mmol/L}$$

$$\text{absorbed phenol} = 2.0 \text{ mmol/g}$$

$$\text{absorbed \%} = 99.68\%$$

$$\% \text{ removed} = \frac{2 \text{ mmol/g} \cdot 5g}{10.63 \text{ g mmol}}$$

$$10.63 \text{ g mmol}$$

$$\% \text{ removed} = 94\%$$

6.26

$$\Delta H + \Delta KE + \Delta PE = Q + W$$

$$1 \text{ g mol H}_2\text{O} = 18 \text{ g/mol}$$

a) From 0 to 100 km/h

$$\Delta KE = \frac{m}{2}(V_2^2 - V_1^2) = \frac{18}{2}(100^2 - 0^2) = 9 \cdot 10000 = 90,000$$

$$(90,000 \frac{\text{km}^2}{\text{s}^2}) \cdot \left(\frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{10^6 \text{ m}^2}{\text{km}^2} \cdot \frac{\text{s}^2}{3600^2} \cdot \frac{\text{J}}{\text{kg m}^2/\text{s}^2} \right) = \boxed{6.94 \text{ J}}$$

b) rising 100 m relates to ΔPE

$$\Delta PE = m \cdot g(h_2 - h_1) = (18 \text{ g})(9.8 \text{ m/s}^2)(100 \text{ m} - 0 \text{ m})$$

$$(17,640 \text{ g m}^2/\text{s}^2) \cdot \left(\frac{1 \text{ kg}}{1000 \text{ g}} \cdot \frac{\text{J}}{\text{kg m}^2/\text{s}^2} \right)$$

$$= 17.64 \text{ J}$$

c) pressure 100° from 0.1

$$\Delta U = \rho \cdot \Delta V = 18 \rho \text{ H}_2\text{O} (2906.2 - 2515.5) \text{ J}/\rho = \boxed{-176.4 \text{ J}}$$

d) Increasing temp feed 0° to 100° @ 5 bar

$$\Delta U = n \cdot C_v \cdot \Delta T = 1 \text{ g mol} \cdot (\text{heat capable of mass water}) \times (100)$$

$$= 1 \text{ g mol} \cdot (4.186 \text{ J/g cm}^3 \cdot 18 \text{ g}) \times (100^\circ \text{C})$$

$$= 75.348 \times 100 = \boxed{7534.8 \text{ J}}$$

e) dissociations to $\text{H}_2 + \text{O}_2$ at 25°C

$$\Delta \hat{U}_r = \Delta H'_r - RT \Delta n_r = (284.84 \frac{\text{kJ}}{\text{g mol}})(1000 \text{ J/kg}) - (8.314 \text{ J/g mol} \cdot \text{K})(298 \text{ K})(0.99 \text{ mol})$$

$$\Delta \hat{U}_r = 284,600 \text{ J}$$

f) undergoing nuclear reaction where all mass converts to energy

$$E = mc^2$$

$$= (18 \text{ g})(2.99792 \times 10^8 \text{ m/s})^2$$

$$= 161.775 \times 10^{18} \text{ g m}^2/\text{s}^2$$

$$= 161.775 \text{ g m}^2/\text{s}^2 \cdot \left(\frac{1 \text{ kg}}{1000 \text{ g}} \right) \times \left(\frac{\text{J}}{\text{kg m}^2/\text{s}^2} \right)$$

$$= \boxed{1.61775 \times 10^{15} \text{ J}}$$